

The Path for LLMs to Acquire Symbolic Logic Capabilities

: the Paradigm Transition from Frequentist to Bayesian

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Abstract: This paper discusses a path to develop logical capabilities in LLMs directly, without adding a symbolic logic deduction layer: directly using the existing phenomenological regularities comprising large amounts of if_then_or statements and content from mathematical derivations or theorems to generate logical corpora for iterative training. This includes training the recognition and classification of condition components within IF clauses. I believe that through this, LLMs will gradually acquire genuine symbolic deductive logic capabilities.

Keywords: Bayesian paradigm, LLM, symbolic logic, IF-THEN-OR

Actually, human logic is essentially a chain or tree diagram of IF THEN, OR statements. It is fundamentally semantic. The criteria and mechanism system for logic originate from semantics. Therefore, in principle, LLMs should also be able to acquire logical capabilities at least at the human level, because the principle is the same. The hallucination problem can be significantly reduced by placing phenomenological data in memory, not participating in the dynamic generation of parallel vector computations on the GPU. Through repeated training with IF statements, hallucinations will be further reduced. Internal logical consistency, the accuracy in recognizing combinations of criteria in IF statements, and the understanding of the logical mechanism between IF and THEN will reach the level of a PhD student in STEM disciplines.

In fact, obtaining an IF THEN training set is very straightforward: decompose and parse the entire content of K-12 textbooks into IF THEN statements, then extend this to higher-level textbooks, and ultimately decompose all current reliable human knowledge into a tree-structured database of IF THEN OR statements, synthesizing it into a training set to train the LLM.

In reality, the essence of most existing symbolic computation libraries is also first compiled into IF THEN, then symbolized. Thus, another direction can be discovered: decompiling existing open-source symbolic libraries, within the scope of their license agreements, into an IF-THEN system.

I believe that continuous training and optimization along these two directions, with retraining, will enable large LLM models to acquire doctoral-level logical capabilities within 3 years.

On this basis, the training set composed of IF-THEN statement libraries can be easily converted into code. This allows for the proposal of a code generation verification layer: by generating equivalent code for a logic and verifying whether the code can run successfully to check logical self-consistency. Code verification is a strong verification method that can meet industrial-grade requirements.

Incorporating the IF THEN training set for training is about upgrading the LLM from a frequency paradigm to a Bayesian paradigm.

The initial phase should focus on building a sophisticated, AI-driven pipeline—a "knowledge compiler." Its purpose is to automate the large-scale conversion of knowledge, from structured data and semi-structured text to natural language descriptions, into a normalized representation of "IF-THEN-OR" logic trees through advanced semantic parsing. The road won't be that smooth, because large portions of human knowledge corpora contain ill-defined concepts and loosely-structured criteria.

This paper presents only a conceptual blueprint; the author currently lacks the resources for preliminary experiments. However, the AI community could readily validate the approach through low-cost, rapid prototyping.

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LLM 获得符号逻辑能力的路径

: 从频率到贝叶斯的范式转移

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摘要：本文讨论不通过加入符号逻辑演绎层，直接通过 LLM 发展逻辑能力的路径：直接用大量 if, then, or 语句的现有现象规律和数学推导或数学定理的内容生成逻辑语料，进行反复迭代训练，包括训练 IF 中的条件组件识别，分类等，我觉得 LLM 会逐步获得真正的符号演绎逻辑能力。

关键词：Bayesian paradigm, LLM, symbolic logic, IF-THEN

其实人类的逻辑就是一个 IF THEN, OR 语句链, 树状图, 本质上是语义的, 逻辑的判据, 机制系统来源于语义, 那么从原理上 LLM 就也能获得至少达到人类水平的逻辑能力, 因为原理是一样的。幻觉问题, 很简单把现象数据放到内存不参与显卡并行向量运算的动态生成, 幻觉将大幅压缩, 通过 IF 语句的反复训练, 幻觉将进一步减少, 内在逻辑一致性, 对 IF 的判据组合的识别准确度, 对 IF 和 THEN 之间的逻辑机制的理解度都会达到 STEM 科学学科的博士生水平。

IF THEN 训练集也不难获得, 就是将 K12 教材的内容全部分解解析为 IF THEN 语句, 然后扩展到更高层次教材, 乃至把所有当前可靠的人类知识全部分解为 IF THEN OR 语句树状数据库, 合成训练集, 训练 LLM。

实际上, 现有大多数符号运算库的本质也是先编译成 IFTHEN, 再符号化。那么可以发现另一个方向是将现有开源符号库在授权协议授权许可的范围内, 反编译成 IF-THEN 体系。

我相信这两个方向的不断训练, 优化, 再训练, 3 年内, 可以让 LLM 大模型习得博士级逻辑能力。

在此基础上的由 IF-THEN 语句库组成的训练集, 是很容易转换为代码, 这里就可以提出一个代码生成验证层, 通过生成一套逻辑的等价代码, 并通过代码能否跑通来验证逻辑自洽性。代码验证是强验证, 可逼近达到工业级需求。

加入 IF THEN 训练集训练, 就是使 LLM 从频率范式升级到贝叶斯范式。

初期, 可考虑开发一个高度智能化的“知识转码”平台或智能化编译器, 它能像工业设备一样, 自动地、批量地将结构化和半结构化的知识(乃至自然语言描述的原理)转换成标准化的“IF-THEN-OR”逻辑树。相关工作不会那么容易, 因为人类知识库语料中有大量定义不清晰, 判据结构不精确的论述。

本文仅提供一份概念性蓝图。受限于资源, 作者尚无力开展前期实验, 但 AI 社区完全可借助低成本方案对该路径进行快速验证。

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